Restaurant Sales Performance Analysis Report

Final Project Report

Prepared By

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Tools Used

- Google Sheets
- Tableau

1. Executive Summary

For this project, I analyzed restaurant sales performance data to identify trends, evaluate operational performance, and develop business recommendations using Tableau.

The goal of the dashboard was to turn raw sales data into a clear and interactive business intelligence tool that could help stakeholders better understand revenue performance, customer behavior, cuisine popularity, and operational trends.

Throughout the project, I used Google Sheets for data cleaning and preparation and Tableau for visualization and dashboard development. The dashboard was designed to provide both high-level KPI summaries and deeper analytical insights through interactive charts and filters.

Some of the most important findings from the analysis include:

- Total revenue exceeded $5 million.
- North Indian cuisine generated the highest revenue among cuisine categories.
- Tirupati was the top-performing city by restaurant revenue.
- Friday generated the highest weekday revenue.
- Revenue trends fluctuated over time and showed signs of decline in later periods.
- Domino’s Pizza ranked as the highest-performing restaurant by revenue.

Based on these findings, recommendations were developed to help improve operational efficiency, increase revenue opportunities, and support future business decision-making.

2. Project Objective

The objective of this project was to evaluate how restaurant sales performance changes over time and identify the primary factors influencing revenue generation.

The final deliverables for the project included:

- An interactive Tableau dashboard
- A report summarizing key findings and business insights
- Strategic recommendations supported by data analysis

The dashboard and analysis were developed to answer the following business questions:

- How has total revenue changed over time?
- Which restaurants generate the highest revenue?
- Which cities contribute the most sales?
- Which cuisines generate the highest revenue?
- Is revenue concentrated among a small number of restaurants?
- Do higher-rated restaurants generate higher revenue?
- Which periods generate the highest order volume?
- What is the average order value over time?

The analysis was also guided by the following hypotheses:

- Revenue increases during weekends and peak dining periods.
- A small number of restaurants generate the majority of total revenue.
- Higher-rated restaurants generate higher overall sales revenue.

3. Data Preparation and Cleaning

Before creating the dashboard, the dataset underwent extensive cleaning and preparation in Google Sheets.

Data Cleaning Tasks Performed

The dataset required several preparation and standardization steps before analysis could begin.

The following data cleaning tasks were completed:

- Removed duplicate records
- Standardized city and restaurant naming conventions
- Corrected inconsistent capitalization

- Verified relationships between datasets
- Checked for missing values and incomplete records
- Converted revenue fields into standardized numeric formats
- Converted mixed INR and USD currency values into USD for consistency
- Applied revenue thresholds to focus analysis on restaurants generating at least $1,000 USD in revenue
- Created calculated metrics for business analysis
- Cleaned address formatting and categorical inconsistencies

Because the original dataset contained both INR and USD currency values, all revenue fields were converted into USD to create a consistent analytical standard across the dashboard.

Additionally, the dashboard analysis focused only on restaurants generating at least $1,000 USD in revenue. This filtering process helped reduce noise from low-performing records and allowed the analysis to focus on more operationally significant businesses.

Calculated Metrics Created

Several calculated fields were developed to support dashboard analysis:

- Total Revenue
- Average Order Value
- Revenue Per Restaurant
- Revenue by Cuisine
- Revenue by City
- Order Count
- Average Rating

These calculations enabled deeper analysis and improved the quality of business insights produced in Tableau.

4. Dashboard Design Process

The dashboard was designed with a focus on simplicity, readability, and business storytelling.

Instead of overcrowding the dashboard with too many visualizations, the goal was to create a layout that allows users to quickly understand performance metrics and identify trends without confusion.

The dashboard includes:

- KPI summary cards for quick business insights
- Revenue analysis by city and cuisine

- Monthly revenue trend analysis
- Weekday sales analysis
- Top-performing restaurant analysis
- Interactive filters and parameters for deeper exploration

Special attention was given to:

- Consistent formatting
- Executive-style dashboard design
- Interactive usability
- Logical chart organization
- Clear labeling and tooltip customization

The overall layout was created to guide users from high-level KPIs into more detailed operational insights.

5. Dashboard Components and Analysis

KPI Summary Cards

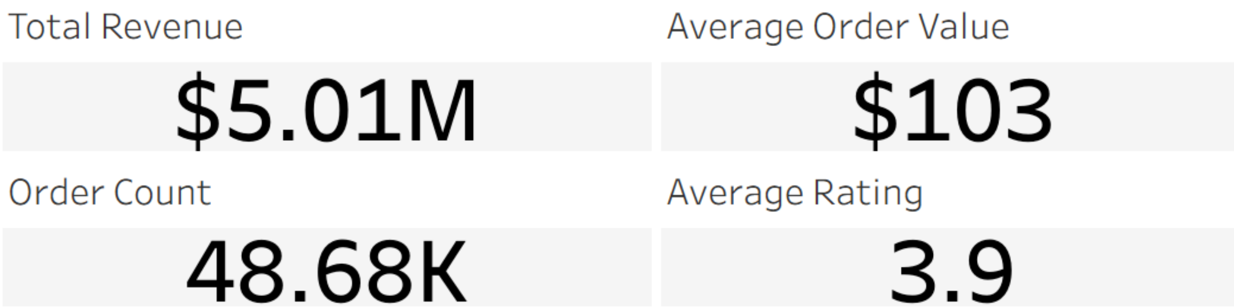
The dashboard begins with four KPI summary cards designed to provide a quick overview of business performance.

KPIs Included

- Total Revenue
- Average Order Value
- Order Count
- Average Rating

Key Findings

- Total revenue exceeded $5 million.
- Average order value was approximately $103.
- Total order count exceeded 48,000 orders.
- Average customer rating was 3.9.



Revenue by City

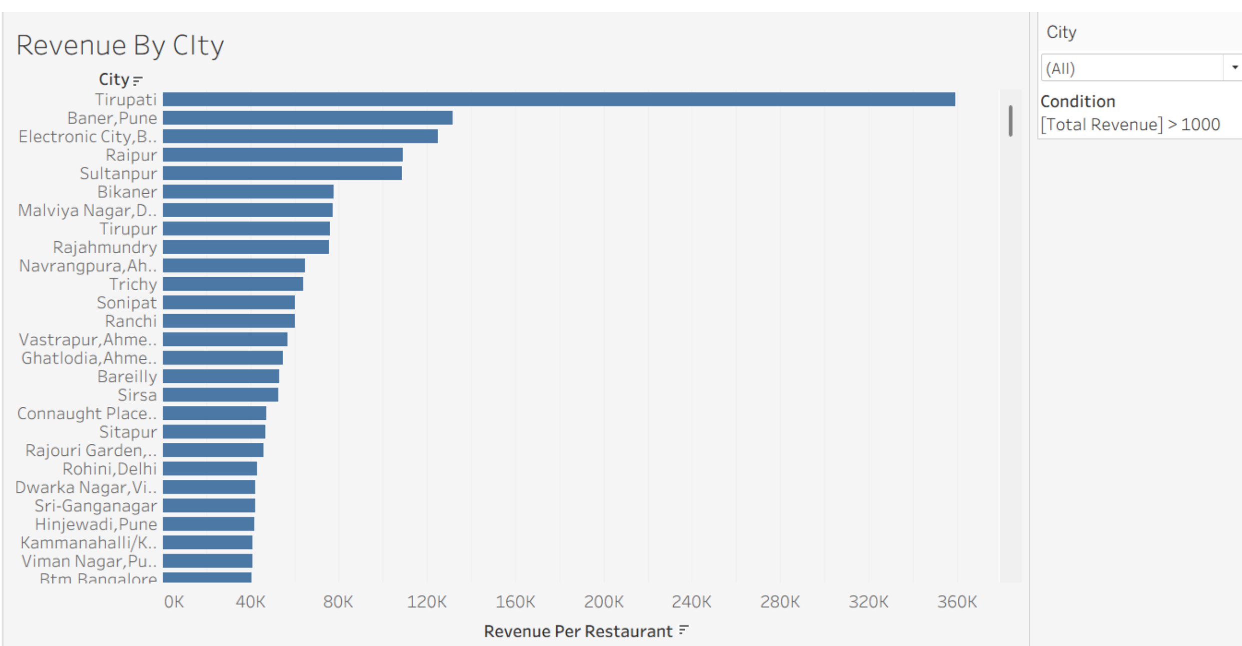
The Revenue by City visualization was created to identify which cities generated the highest sales revenue among restaurants with at least $1,000 USD in total revenue.

Findings

- Tirupati generated the highest restaurant revenue.
- Revenue was heavily concentrated among a smaller number of cities.
- Several cities significantly outperformed the rest of the dataset.

Business Insight

The analysis suggests the business could benefit from focusing operational and marketing efforts in high-performing regions.



Revenue by Cuisine

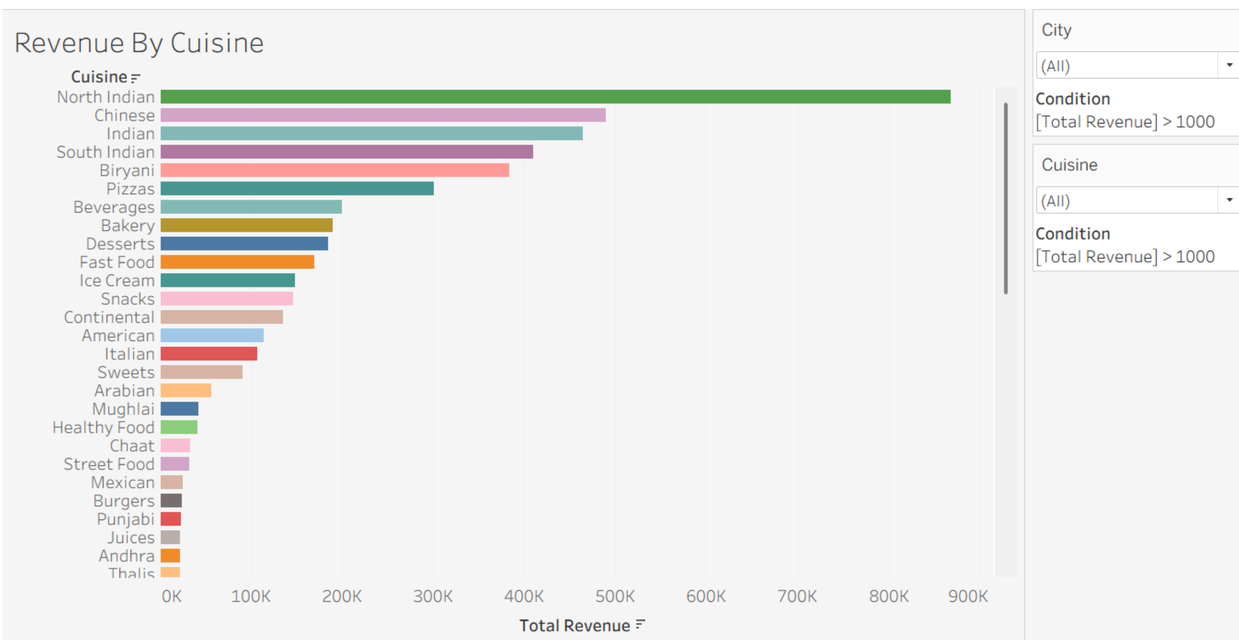
The Revenue by Cuisine visualization was designed to identify which cuisine categories generated the highest revenue after standardizing all revenue into USD.

Findings

- North Indian cuisine generated the highest revenue.
- Chinese and Indian cuisines also performed strongly.
- Some cuisine categories generated significantly lower revenue than others.

Business Insight

Customer demand appears to be concentrated among a smaller number of cuisine categories, which could help guide future menu and marketing decisions.



Monthly Revenue Trend Analysis

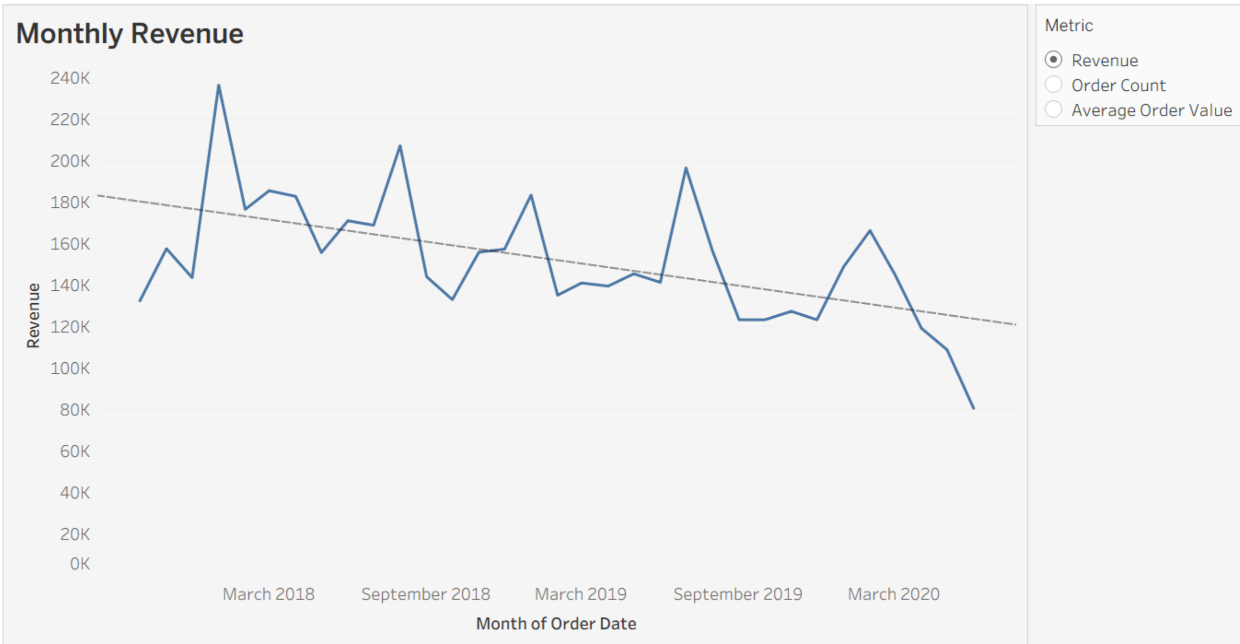
The Monthly Revenue chart was created to evaluate how sales performance changed over time.

Findings

- Revenue fluctuated throughout the dataset.
- Multiple months showed strong spikes in revenue.
- The trend line showed a gradual decline over later periods.

Business Insight

The business may need to investigate operational or market-related factors contributing to declining revenue trends.



Revenue by Weekday

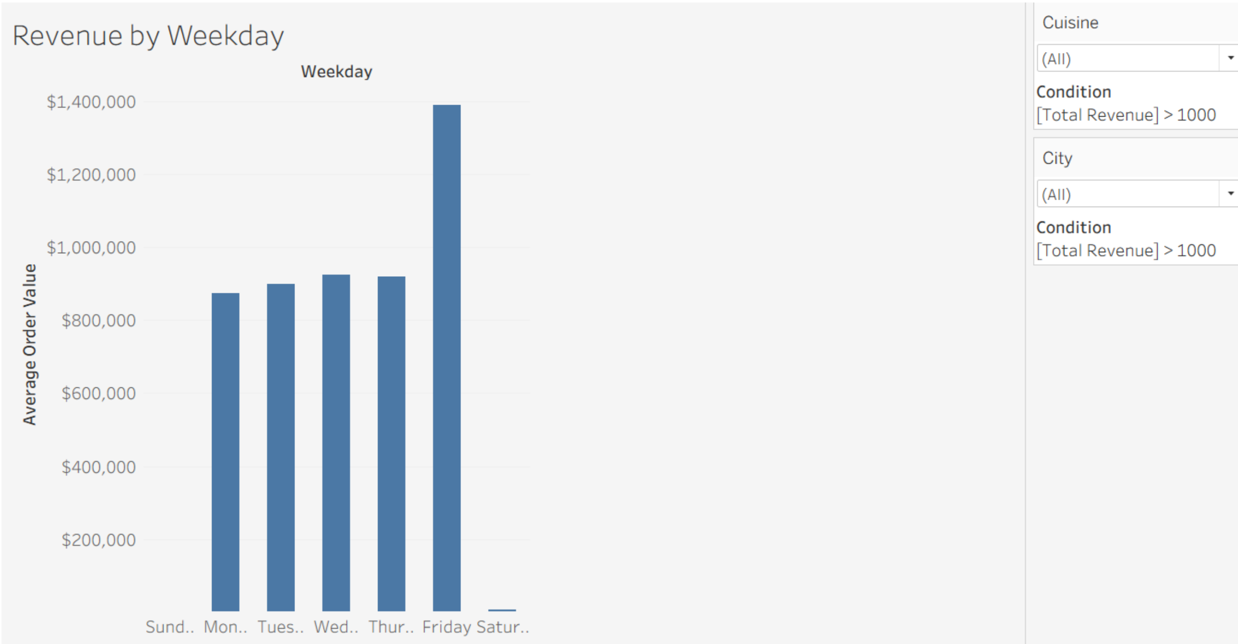
The Revenue by Weekday visualization was used to identify which periods generated the highest sales activity.

Findings

- Friday generated the highest revenue.
- Monday through Thursday remained relatively consistent.
- Saturday revenue was significantly lower than expected.

Business Insight

These findings suggest customer demand may peak before weekends rather than during weekends.



Top Restaurants Analysis

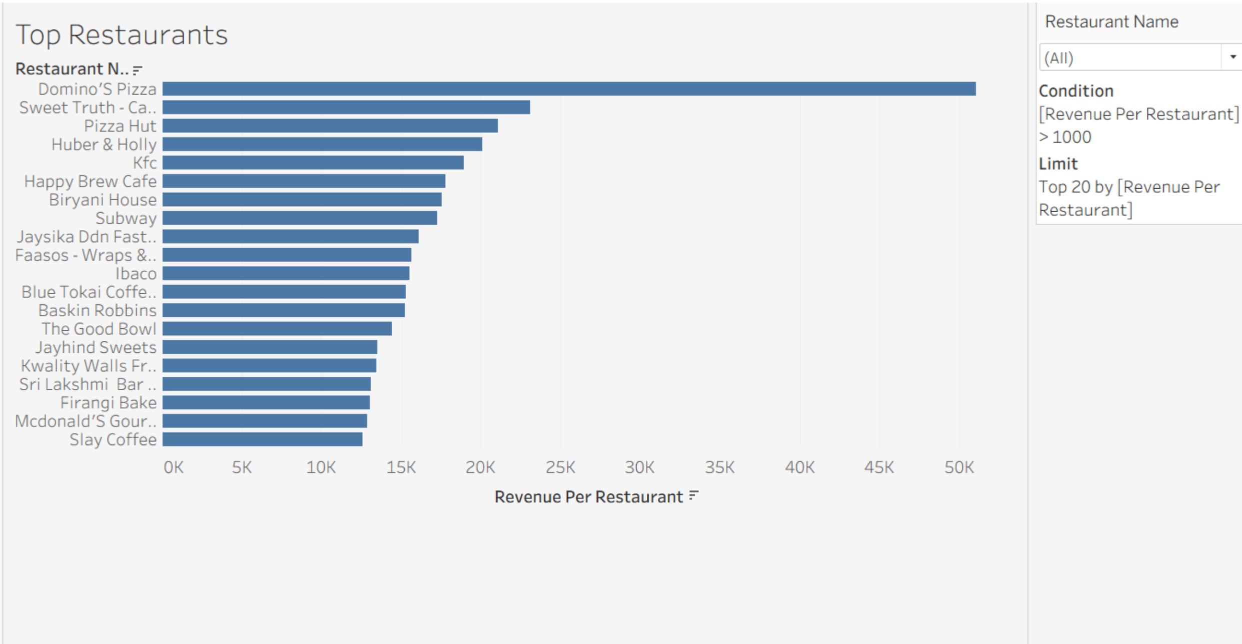
The Top Restaurants visualization identified restaurants generating the highest overall revenue.

Findings

- Domino's Pizza generated the highest revenue.
- A relatively small number of restaurants generated a large portion of total sales.
- Several brands significantly outperformed competitors.

Business Insight

The business may benefit from studying the operational strategies of top-performing restaurants and applying similar practices elsewhere.



6. Dashboard Interactivity Features

Interactive filters were added to improve user experience and allow deeper exploration of the data.

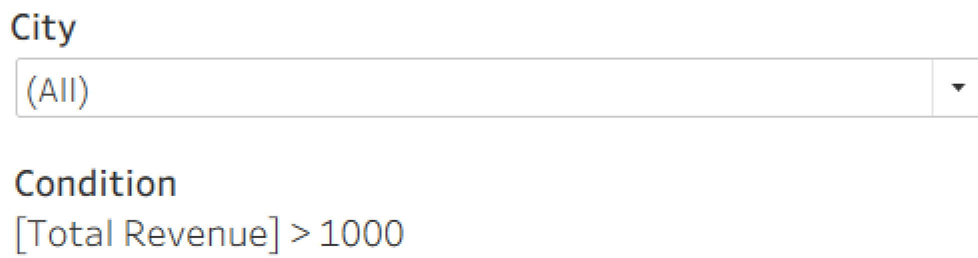
Filters Included

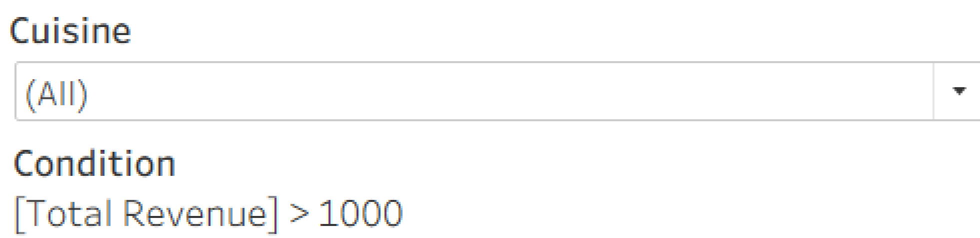
- City Filter
- Cuisine Filter
- Restaurant Filter
- Metric Selection Parameter

These filters allow users to dynamically customize analysis and explore relationships between variables.

Benefits of Interactivity

- Improved executive usability
- Faster insight discovery
- Increased dashboard flexibility
- Better user engagement
- Enhanced analytical exploration





7. Hypothesis Evaluation and Business Conclusions

Hypothesis 1 — Revenue Increases During Weekends and Peak Dining Periods

This hypothesis was partially supported by the analysis. Friday generated the highest revenue overall; however, Saturday revenue was significantly lower than expected. This suggests customer demand may peak before the weekend rather than during the weekend itself.

Hypothesis 2 — A Small Number of Restaurants Generate the Majority of Revenue

The analysis strongly supported this hypothesis. Revenue was concentrated among a relatively small number of restaurants, cities, and cuisine categories.

Hypothesis 3 — Higher-Rated Restaurants Generate Higher Revenue

The dashboard included customer ratings as a performance metric; however, additional correlation analysis would be needed to fully determine the relationship between ratings and revenue generation.

Overall Business Conclusions

- Revenue performance varied significantly across cities and restaurants.
- A small number of restaurants and cuisines generated a large percentage of total revenue.
- Revenue trends fluctuated over time and showed signs of decline in later periods.
- Friday generated the strongest operational performance.
- Interactive dashboards improve the ability to identify business trends and operational opportunities.

8. Business Recommendations

The following recommendations were developed based on the analysis findings.

Recommendation 1 — Expand High-Performing Cuisine Offerings

The business should prioritize investment in high-performing cuisine categories such as North Indian and Chinese cuisine.

Recommendation 2 — Increase Focus on Top Cities

Marketing and operational resources should be concentrated in cities demonstrating the strongest revenue performance.

Recommendation 3 — Improve Weekend Revenue

Weekend promotional campaigns and operational adjustments should be implemented to increase Saturday performance.

Recommendation 4 — Investigate Revenue Decline Trends

The business should conduct additional analysis into declining monthly revenue trends to identify underlying causes.

Recommendation 5 — Replicate Top Restaurant Strategies

Successful operational practices from high-performing restaurants should be evaluated and implemented across other locations.

9. Skills and Technologies Demonstrated

This project demonstrated practical skills in:

Technical Skills

- Data Cleaning
- Data Transformation
- Data Validation
- Dashboard Design
- Data Visualization
- Business Intelligence
- Interactive Filtering
- KPI Development
- Calculated Fields
- Trend Analysis
- Business Reporting

Software Skills

- Google Sheets
- Tableau

10. Final Reflection

This project gave me hands-on experience working through the full analytics process, from cleaning raw data to building a professional dashboard and creating business recommendations.

One of the biggest takeaways from this project was learning how important data preparation and dashboard design are when presenting information to stakeholders. Throughout the project, I worked through issues involving data cleaning, calculated fields, formatting, relationships, filtering, and tooltip customization in Tableau.

I also gained a better understanding of how dashboards should tell a story instead of simply displaying charts. Every visualization included in the dashboard was selected to answer a specific business question or support a business recommendation.

Overall, this project helped strengthen my skills in:

- Data cleaning and preparation
- Tableau dashboard development
- Business-focused data visualization
- KPI analysis
- Trend analysis
- Interactive dashboard design
- Communicating analytical findings professionally

The final dashboard successfully transforms raw restaurant sales data into actionable business insights while maintaining a clean and professional presentation style.

11. Dashboard Link

https://public.tableau.com/views/RestaurantSalesPerformaneDashboard/Dashboard1?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

12. Appendix

Additional Dashboard Screenshot